

SMART

REMIND

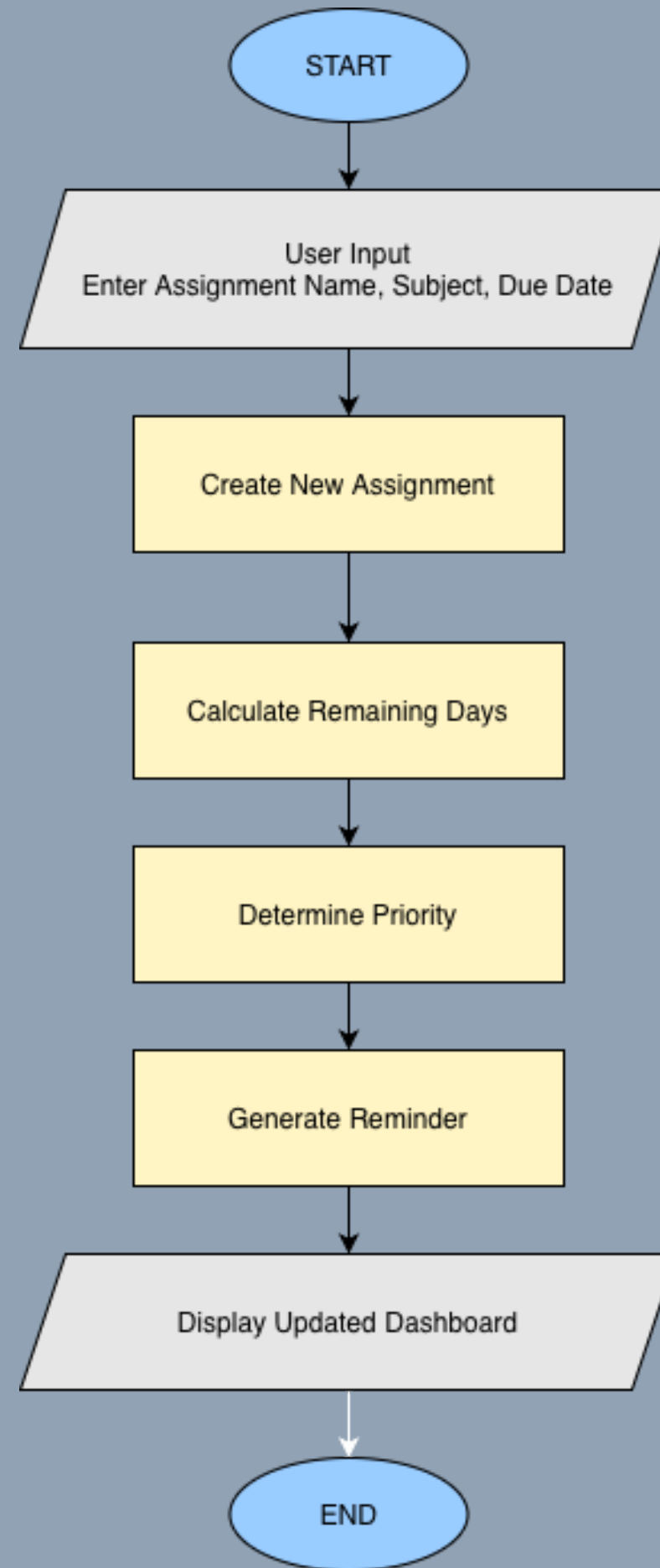
SITI AISYAH 25013517

SMART REMIND

SmartRemind is a Reflex-based Assignment Reminder designed to help students manage academic deadlines more effectively. The system tracks assignments, calculates urgency levels based on due dates, generates reminders, and provides a dashboard for monitoring upcoming and overdue tasks.

The goal of SmartRemind is to reduce missed deadlines and improve student time management through automated deadline awareness and assignment prioritization.

WORKFLOW



TOOL AND LIBRARIES

Tool / Library	Purpose
Python	Main programming language
Reflex	Builds the web application's user interface
Pydantic	Validates and structures data models
JSON	Stores configuration or application data, depending on how your project uses it.
datetime	Built-in Python module for handling dates and times
typing	Built-in Python module for type hints such as List.

CODE

```
import reflex as rx
from datetime import date, datetime
from typing import List
from pydantic import BaseModel

#data
class Assignment(BaseModel):
    id: int
    name: str
    subject: str
    due_date: str
    days_remaining: int = 0
    priority: str = "upcoming"
    reminder: str = ""
    done: bool = False
```

```
#calculate date
def calc_days(due_date_str: str) -> int:
    try:
        due = datetime.strptime(due_date_str, "%Y-%m-%d").date()
        return (due - date.today()).days
    except Exception:
        return 999

#prio
def calc_priority(days: int) -> str:
    if days < 0: return "overdue"
    if days == 0: return "due_today"
    if days == 1: return "due_tomorrow"
    return "upcoming"

#generate reminder
def generate_reminder(name: str, subject: str, days: int, priority: str) -> str:
    if priority == "overdue":
        return f"🚨 OVERDUE by {abs(days)} day(s)! Submit '{name}' immediately and contact your lecturer."
    if priority == "due_today":
        return f"🔴 '{name}' ({subject}) is due TODAY! Stop everything and finish it now."
    if priority == "due_tomorrow":
        return f"🔴 '{name}' ({subject}) is due TOMORROW. Finish it tonight – no more delays!"
    if days <= 3:
        return f"🟡 '{name}' ({subject}) is due in {days} days. Start now to avoid last-minute panic."
    if days <= 7:
        return f"🟡 '{name}' ({subject}) is due in {days} days. Make a plan and start drafting."
    return f"🟢 '{name}' ({subject}) is due in {days} days. You have time."
```


DASHBOARD

SmartRemind

Assignment Reminder

Dashboard & Analytics

3

Total Assignments

0

Overdue

2

Upcoming

New Assignment

Assignment Name

e.g. Lab Report

Subject

e.g. Biology

Due Date

06/14/2026

+ Add Assignment

Filter by subject:

All

Sort by:

due_date

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

AI Agent Report

Artificial Intelligence · 2026-06-14

Due Today



Today!



Data Structures Quiz

Computer Science · 2026-06-16

Upcoming



2 days left



Network Security Essay

Cybersecurity · 2026-06-21

Upcoming



7 days left



Reminders

Generate Reminders

INPUT EXAMPLE

New Assignment

Assignment Name

Week 7 Task

Subject

Open Source

Due Date

06/18/2026

Jun 2026

Su

Mo

Tu

We

Th

Fr

Sa

31

1

2

3

4

5

6

7

8

9

10

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12

13

14

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22

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25

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30

1

2

3

4

5

6

7

8

9

10

11

+ Add Assignment

All

Sort by: due_date

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

AI Agent Report

Artificial Intelligence · 2026-06-14

Due Today

Today!

✓

Data Structures Quiz

Computer Science · 2026-06-16

Upcoming

2 days left

✓

Week 7 Task

Open Source · 2026-06-18

Upcoming

4 days left

✓

Network Security Essay

Cybersecurity · 2026-06-21

Upcoming

7 days left

✓

Reminders

Generate Reminders

'AI Agent Report' (Artificial Intelligence) is due TODAY! Stop everything and finish it now.

'Data Structures Quiz' (Computer Science) is due in 2 days. Start now to avoid last-minute panic.

'Week 7 Task' (Open Source) is due in 4 days. Make a plan and start drafting.

'Network Security Essay' (Cybersecurity) is due in 7 days. Make a plan and start drafting.

Filter by subject

due_date

✓ All

Artificial Intelligence

Computer Science

Cybersecurity

Open Source

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

AI Agent Report

Artificial Intelligence · 2026-06-14

Due Today

Today!

✓

Data Structures Quiz

Computer Science · 2026-06-16

Upcoming

2 days left

✓

Week 7 Task

Open Source · 2026-06-18

Upcoming

4 days left

✓

Network Security Essay

Cybersecurity · 2026-06-21

Upcoming

7 days left

✓

INPUT EXAMPLE

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

- 📅

Week 7 Task

📅 Open Source · 📅 2026-06-18

Upcoming

4 days left

✓

🗑️
- 📅

Network Security Essay

📅 Cybersecurity · 📅 2026-06-21

Upcoming

7 days left

✓

🗑️

Reminders

🔔 Generate Reminders

- 🕒

'Week 7 Task' (Open Source) is due in 4 days. Make a plan and start drafting.
- 🕒

'Network Security Essay' (Cybersecurity) is due in 7 days. Make a plan and start drafting.

Completed Assignments

- 🔍

AI Agent Report

Artificial Intelligence

🕒

🗑️
- 🔍

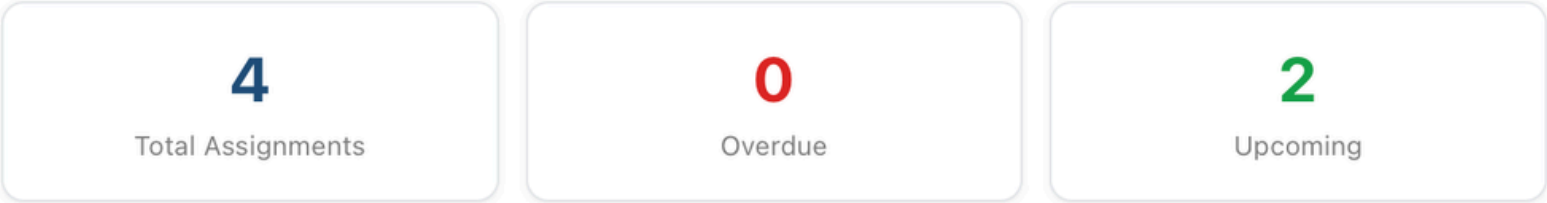
Data Structures Quiz

Computer Science

🕒

🗑️

Dashboard & Analytics



Thank you!